ITW-Hobart

Hands Free Pager Device

Software Design Specs

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1 OVERVIEW

2 INTRODUCTION

This document is the designer's guide for the ITW-Hobart ***Hands Free Pager Device (HFPD)***, an electronic device that allows hands free unlocking of equipment and the receipt of status, error and other types of messages for employees via Bluetooth communications. The purpose of this document is to detail the various design criteria for the hands free pager device for the software developers of this product and the other products that must interface with this device.

3 BACKGROUND

The HFPD was designed to increase employee and workplace productivity by automating the process of unlocking and locking access to equipment, and to inform employees of equipment issues. This functionality is achieved via Bluetooth wireless communications between the HFPD and the various pieces of equipment that the employee will use. The HFPD is powered by batteries so its design must be energy efficient and use low power mode whenever possible.

4 HARDWARE

The HFPD contains an ePaper display configured into three regions:

- Top left corner is a section dedicated to showing a graphic image, a logo,
- Top right corner is a section which contains the employee name, 2 lines of 10 characters each, and
- The bottom portion is capable of displaying alert messages, 2 lines of 16 characters each.

The ePaper display does not require any power to maintain its display, only to change the data displayed. Once the display has been changed with the proper logo, employee name and alert message, that information stays on the display without any additional power.

The HFPD has a bi-color red/green LED for showing a heartbeat, activity and status. There is also a beeper with a fixed frequency and loudness that can be used to alert the employee when something happens, like an alert message is received. And there is a single recessed push button which is used to turn the unit on/off and clear alerts.

5 MODES OF OPERATION

6 AWAY MODE

To maximize battery life, the HFPD should use the lowest possible power mode whenever possible. If we assume that the typical employee using this device works 40 hours per week, then the device only needs to be on and functional for approximately 40/168 hours per week, so there is the potential for significant energy savings when not in use.

The HFPD does not come with a mechanical switch to disconnect power when not in use, so its microcontroller will power off and/or put into sleep mode as much circuitry as possible when idle, not in use. We call this mode of operation the AWAY mode, which represents the mode when the unit is not in the workplace and in use by the employee. In this mode there are no Bluetooth communications or other activity, the microcontroller is in its power saving mode waiting for a button press to wake up the device.

7 SHIFT MODE

When not in the AWAY mode, the HFPD is in SHIFT mode of operation where the employee is working, on shift. In this mode the HFPD will have its Bluetooth communications active, sending out advertisements and awaiting for alerts and other messages. This mode of operation starts with a button press on the device to wake it up from its AWAY mode.

Upon starting this mode of operation, the HFPD initializes a countdown timer with a programmable interval which represents the shift time, how long the HFPD should stay in this mode. Once this timer expires, the HFPD will switch over to AWAY mode for power savings.

8 BLUETOOTH OPERATIONAL DETAILS

9 ADVERTISEMENTS AND COMMUNICATIONS

The HFPD announces its presence via BLE advertisements when operating in the SHIFT mode. These advertisements serve multiple purposes which include notifying the Bluetooth enabled equipment of an employee's presence to unlock that equipment and to receive messages from the equipment for configuration and service issues.

While in SHIFT mode, the HFPD will spend its time doing advertisements and listening for connections from the equipment, in case there are settings that need to be updated or alerts for service events. After each advertisement transmission there a period of time allotted for listening for a connection. Once this period of time passes, assuming there is no connection made, the HFPD will enter into low power sleep mode until it is time to do the next advertisement.

The dual functionality requirements of unlock and messaging require periodic changes to the advertisements to support both functions.

10 EQUIPMENT UNLOCK/LOCK

The primary function of the HFPD is to enable hands free log-ins for Bluetooth enabled equipment. The employee with their HFPD would just walk up to the equipment and that equipment would sense their presence and if permitted, automatically unlock the equipment for their use while that user is present at the machine.

In order for this functionality to succeed, the equipment to be unlocked only want to sense HFPD that are within 2-3 feet of the unit, not devices on the other side of the room. The best method to enable this functionality is to reduce the transmit power of the HFPD to its minimum setting, which will prevent HFPD outside this 2-3 feet radius from being seen. In the event that multiple HFPD are seen, the equipment will need to have some logic for deciding which employee is at the user interface and waiting for it to unlock. This logic could be first seen combined with the signal strength as seen by the equipment of all HFPD seen, or something else. Once logged in, that employee should stay logged in until they walk away and their advertisements are no longer seen, the signal is too weak to be seen by that piece of equipment.

11 EQUIPMENT SERVICE MESSAGES

A secondary function of the HFPD is to provide a mechanism for the Bluetooth enabled equipment to inform employees that there is an issue, a service event. This could be anything like out of paper or some other event. In order to send messages to the HFPD, the equipment must wait for an advertisement from that HFPD and then promptly connect in the allotted time slot to send its message.

This function has a different requirement for the Bluetooth communications in that the HFPD should be transmitting at its maximum power setting for the longest range possible, so a HFPD far away from the equipment can still be seen and as such, receive the alert message.

12 DUAL ADVERTISEMENT MODES

The primary and secondary functions contradict each other for the required transmit power setting. In order for the unlock function to work, we need minimal transmit power on the HFPD so only those devices very close are seen, whereas maximum transmit power on the HFPD is preferred to receive alerts at maximum range.

In order to accommodate both functions, the HFPD must switch between minimum and maximum transmit power when advertising. However that alone is not sufficient since a HFPD that is further than 2-3 feet away may appear to be very close to the equipment if it is transmitting at its maximum power setting, causing the equipment to unlock erroneously. To prevent this from happening, the data used in the advertisement must differ in the low and high transmit power modes, so the equipment can determine if the seen HFPD is actually close or not. The equipment should only unlock if the seen advertisement was sent in low power mode. This only applies to the unlock function, the message functionality does not care if the HFPD is close or far away, so any seen advertisement is suitable for connecting and sending the messages.

13 ADVERTISEMENT DATA

Advertisements need to be processed by the Bluetooth enabled equipment to identify employees to unlock access and send messages. These advertisements only contain a minimal amount of data, which includes:

- its device address (its MAC address, manufacturer data, after manufacturer ID),
- its flags (if HFPD is discoverable or not, etc.),
- its transmit power level (-20 = minimum power, need for unlock),(4dBm = max power)
- its battery service characteristic (battery life remaining in 0-100 percent), and
- its local name (ITWH)
- Eg:.
- [0x01] 06 Flags = GeneralDiscoverable, BrEdrNotSupported
- [0x0a] ec Tx power -20 dB
- [0xff] ffffb0d6e809379 manufacturer ID FFFF + MAC address as requested
- [0x16] 0f1864 battery = 0x64 = 100 dec percent
- [0x09] 49545748 name = ITWH (had to shorten, MAC address addition over firmware version meant less bytes in advert)
-

Equipment should first verify that the seen advertisement has the correct local name before continuing to process it. If a match, it should then check the MAC address against its internal database to see if this is a known employee. If the employee is known, the transmit power field is checked next.

If the transmit power is at its minimum setting of -20, and that user is permitted to unlock the user interface which is still locked, proceed with unlock.

If there are any messages for that employee, the transmit power level does not need to be checked, just connect and exchange the messages as required.

14 MESSAGES

After a Bluetooth enabled piece of equipment has seen an advertisement from the HFPD and it has determined that it needs to send it a message, it must connect within the allotted time slot immediately following the advertisement, as per the Bluetooth spec. Once connected, the HFPD is ready to receive messages. When all messages have been sent, the equipment should disconnect from the HFPD, letting the HFPD return to its normal operation. Supported messages are shown in table 3.6.1. All messages send data in ASCII format, not binary so if sending a value for 60, the message payload should contain a 6, followed by a 0, followed by a space.

Base 128-bit UUID = 8F90xxxx-B457-4761-8F7A-07F1AEBCB453

UUID	Characteristics	Message Name	Formatting	Notes
0001	Open Read Open Write	Employee Name Top Line	10 ASCII characters (20h – 75h)	Send exactly 10 characters per line, no more, no less. Use spaces as necessary to fill.
0002	Open Read Open Write	Employee Name Bottom Line	10 ASCII characters (20h – 75h)	
0011	Open Read Open Write	Alert – Top Line	16 ASCII characters (20h – 75h)	Send exactly 16 characters per line, no more, no less. Use spaces to fill. Beeper sounds with bottom line update.
0012	Open Read Open Write	Alert – Bottom Line	16 ASCII characters (20h – 75h)	
0018	Open Write	Alert – Clear	< ignored, send any byte>	Clears the current alert
0020	Open Read Open Write	Beeper Configuration	# of milliseconds (ASCII, terminate with a space)	How long beeper sound with alert (0 = no sound)
0030	Open Read Open Write	Graphic Image	Index of image to use (ASCII, 0=none, 1=ITW Hobart)	Select image from programmed images
0040	Open Read Open Write	Shift Duration	Hours,Minutes (ASCII 0-9 for each value, terminate minutes with space)	How long to stay on after being awakened with button press
0050	Open Read Open Write	Advertisement Configuration	# low power, msec period (ASCII 0-9 for each value, terminate delay with space)	How many low power advertisements to send back to back before sending a single high power one, the delay between adjacent low power advertisements. *Note that there is a 1 second resolution on timeouts so the period should be in 2000 ms

				or multiples to avoid rounding and truncation issues.
3550	Open Read	Firmware Version	ASCII text version string	For ex. “V0.1”

Table 3.6.1

We use Nordic Semiconductor IWTH BLE. We can receive data like below.

```
if (p_evt_write->uuid.uuid == ITWH_NAME_TOP_CHAR_UUID)
   {
      memcpy(nvSettings.nameFirst, p_evt_write->data, 10);
}
```

15 EMPLOYEE NAME

Data sent via this message will be used to show the employee’s name on the ePaper display. Once received, this data is stored in non-volatile flash to be kept across power loss and reboot events. The sender has to send 10 characters for each line, so if the name is shorter than 10 characters, use spaces to fill in the remaining unused characters.

16 ALERTS

Data sent via this message will be used to show an alert on the ePaper display. This data is not stored in non-volatile flash, so it can be lost. The sender has to send 16 characters for each line, so if the string to show is shorter than 16 characters, use spaces to fill in the remaining unused characters. Alerts will trigger the beeper to sound, if the beeper has been enabled. Alerts can be cleared via the alert clear message or via the employee pressing the button.

17 BEEPER CONFIGURATION

This message will configure the beeper usage, how long the beeper should sound (in milliseconds) when it is requested. To disable the beeper, set this value to zero. This setting is saved to non-volatile flash. It has a default value of 2 seconds.

18 GRAPHIC IMAGE

This message will be used to select a pre-programmed image to display in the logo area of the ePaper display. Once received and if this index is valid, it is stored in non-volatile flash to be kept across power loss and reboot events. The default image shown is index 0 which is the ITW Hobart logo.

19 SHIFT DURATION

This message will be used to set a timer for how long the unit should stay on after being awakened, before it turns itself off to save battery power. The amount of time specified here should be long enough to cover the employee’s shift once the unit has been turned on at shift start. This data is stored in non-volatile flash to be kept across power loss and reboot events. It has a default setting of 8 hours, 40 minutes.

20 ADVERTISEMENT CONFIGURATION

This message will be used to define how the HFPD operates, how quickly it can unlock a piece of equipment and how much time it can spend in low power mode to minimize battery usage. When in SHIFT mode, the HFPD will continuously send out advertisements to notify Bluetooth enabled equipment of its presence. This advertisement event consists of a transmission of the HFPD data, followed by waiting for any possible connection, and then a period of dormancy where it goes into low power mode (asleep, to save battery usage) waiting to repeat the process all over again.

The period (2nd) setting represents how frequent these advertisements should occur, every XXX milliseconds. The smaller the value, the quicker the equipment will be able to see the HFPD and unlock for the employee to use. However shorter delays mean less time in sleep mode, resulting in more battery power being consumed and a shorter battery life. The default period setting is 2000 msec.

The number of low power advertisements (1st) setting defines how many of these unlock advertisements to do consecutively before the HFPD switches over to do a single high power advertisement. This is required to support the receipt of messages when not in front of the equipment, further than ~3 feet away. Smaller values equate to quicker response times, when the remote employee will be able to receive alert messages. However smaller values also mean more power is used to do the high power advertisements. The default value for this setting is 10, meaning if there is no employee at the equipment when the alert is generated, they should be notified after 10 x period = 10 x 2000 msec = 20 seconds.

Both of these settings are stored in non-volatile flash. The high power advertisement should be executed immediately after the last low power advertisement has finished waiting for a connection event and before that sleep period.

21 USAGE CONFIGURATION

22 BUTTON

The button serves multiple purposes on the HFPD, so how to handle depends on when it is pressed.

If in AWAY mode, the button is used to awaken the unit and place it in SHIFT mode~~. To avoid accidental activations, the button needs to be pressed and held down for at least 1 second to switch into SHIFT mode.~~

If in SHIFT mode, the button can be used to clear alerts and prematurely end SHIFT mode, go into AWAY mode. The duration of the button press indicates which function is being requested by the employee. To clear an alert from the screen, press the button and then release. If you press and hold the button down for at least 5 seconds, this will turn off the HFPD and put it into AWAY mode.

23 LED

The bi-color red/green LED is used to show periodic status when in SHIFT mode. When in AWAY mode, this LED should be off, never blinks.

When the HFPD switches into SHIFT mode, this LED will briefly turn green for every high power Tx, after which it will be off for the majority of the time. It should blink with every high power advertisement, so using default settings what would be every 20 seconds. ~~The color used indicates the battery status, with green being battery is okay, and red being low battery, should change soon.~~ A brief green blink when clearing an alert. A brief red blink when holding down longer than 5 seconds to turn off.

24 REVISION HISTORY

25 REVISION X0

Initial release.

26 REVISION X1

Updated for characteristic UUID and more details.

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